

Forum

Unanticipated specificity
in effector-triggered
immunity

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Effector-triggered immunity (ETI) enables hosts to react to pathogens by monitoring few key cellular processes. ETI responses are assumed to be similar toward related pathogen effectors. However, recent evidence from the invertebrate model *Caenorhabditis elegans* and pore-forming toxins indicates a much more complex and specific ETI than previously anticipated.

Innate immune recognition is generally considered to be unspecific or semispecific, allowing distinction of different types of pathogens (e.g., Gram-positive versus Gram-negative pathogens), but not of different strains of the same pathogen species. **Pattern recognition receptors (PRRs)** (see [Glossary](#)) mediate the recognition of **microbe-associated molecular patterns (MAMPs)**, which are directly derived from pathogenic as well as commensal microbes and have been studied intensively over the last decades in the innate immune systems of both vertebrate and invertebrate taxa. Recently, the recognition of active virulence-associated processes has been identified as a key mechanism of innate immune activation, which permits the discrimination of pathogens and nonpathogens. Here, innate defense responses are triggered by recognition of endogenous (i.e., host-derived) **damage-associated molecular patterns (DAMPs)** and/or recognition of molecular perturbations that result from

an effector's virulence function, termed **'effector-triggered immunity' (ETI)**.

While specificity in MAMP and DAMP recognition is dependent on the receptor-ligand mechanism and thus on the diversity of germline-encoded innate immune and non-immune receptors and receptor complexes, **ETI relies on the surveillance of key cellular processes, such as translation, mitochondrial processes, or membrane function, that are often targeted by pathogen effectors [1].** Thus, the overwhelming consensus is that the surveillance of these few key cellular processes enables the host to respond to a large variety of different pathogens [2]. Consequently, virulence factors from diverse pathogens, which target common host processes, are considered to have similar functional outcomes, despite being structurally diverse. **For example, pore-forming toxins (PFTs) are virulence factors produced by many pathogenic bacteria that introduce transmembrane pores in the plasma membrane and show a great diversity in structure [3].** PFT-mediated disruption of plasma membrane integrity causes influx of calcium and efflux of potassium. These alterations in the intracellular ion composition activate cellular defense responses, repair pathways, and protective cytoskeletal remodeling [4]. The nucleotide-binding leucine-rich repeat (NLR) family pyrin domain containing protein 3 (**NLRP3**) acts as a 'guard' of cytosolic potassium concentrations. The NLRP3 inflammasome is activated by potassium efflux and mediates ETI by engaging caspase-1 and mediating **pyroptotic cell death [5].** Thus, NLRP3 is regarded as a general sensor of disrupted membrane integrity that can be induced by a diverse array of pathogen effectors [1].

Surprisingly, **invertebrates, which only possess the innate arm of immunity, have been repeatedly shown to mount highly specific cellular responses to bacterial PFTs with the common function of generating pores in host intestinal cell membranes (Figure 1, Key figure) [6–10].** Even though this is

Glossary

Damage-associated molecular patterns (DAMPs):

endogenous molecules released by stressed, damaged, or dying cells. They act as danger signals, alerting the immune system to cell damage and triggering an inflammatory response by interacting with pattern recognition receptors (PRRs) on immune cells.

Effector-triggered immunity (ETI): an immune response activated by sensing pathogen effectors directly or indirectly. Indirect sensing includes the recognition of perturbations in core cellular processes that result from an effector's virulence function.

Microbe-associated molecular patterns (MAMPs):

conserved molecular structures found on specific groups of microbes, such as lipopolysaccharides in bacteria. They are recognized by PRRs in the host immune system, initiating an immune response to eliminate the invading pathogen.

Mitochondrial unfolded protein response:

protective cellular mechanism triggered by the accumulation of misfolded proteins in the mitochondria. The accumulation of misfolded proteins can be triggered by environmental stressors, including pathogens. This response leads to increased expression of chaperones and proteases to restore mitochondrial function and maintain cellular health.

NLRP3: pattern recognition receptor that forms part of the inflammasome, a multiprotein complex responsible for activating inflammatory responses. When triggered by pathogens, DAMPs, or MAMPs, NLRP3 initiates the release of proinflammatory cytokines, such as IL-1 β , leading to inflammation.

Pathogen-associated molecular patterns:

related to MAMPs, but only including molecular patterns unique to pathogens, such as specific toxins, that are then also recognized by PRRs.

Pattern recognition receptors (PRRs): proteins expressed by host cells that recognize conserved molecular patterns (MAMPs, DAMPs, or pathogen-associated molecular patterns) found on commensals, pathogens, or damaged cells. PRRs play a crucial role in the innate immune system by detecting infection and initiating an immune response.

Pyroptotic cell death or pyroptosis: form of inflammatory programmed cell death typically triggered by infection. It is mediated by inflammasomes and caspase-1 activation, leading to the release of proinflammatory cytokines and pore formation in the cell membrane, causing cell lysis, and promoting an immune response.

generally known in the field of insect and nematode immunity research, the implications of this PFT-triggered immune specificity have been overlooked. Here, we use available comprehensive data on the interaction between the nematode host *Caenorhabditis elegans* (Box 1) and its PFT-producing pathogen *Bacillus thuringiensis* (Bt) as a model to

Box 1. *C. elegans* immunity

Lacking an adaptive immune system, *C. elegans* relies solely on innate immunity, with defense responses primarily executed by epithelial cells in the epidermis and intestine (Figure 1). These responses include the production of a variety of antimicrobial proteins, reactive oxygen species, and cellular processes such as autophagy [19].

Due to immune-related similarities with humans, including intestinal cell function and signaling cascades, *C. elegans* serves as a highly informative, experimentally accessible model for studying innate pathogen defenses. Its immunity is regulated by a complex network of conserved signaling pathways, such as those based on p38 and extracellular signal-regulated kinase mitogen-activated protein kinases, transforming growth factor beta, and insulin-like receptor/DAF-2 (reviewed in [20]). *C. elegans* lacks NOD-like receptors. Other putative PRRs, such as C-type lectin receptors and the one nematode Toll-like receptor (TOL-1), have yet to be shown to be involved in pathogen detection. Despite this, *C. elegans* is capable of recognizing bacterial pathogens by sensing bacteria-derived products in the environment, including small RNAs, secondary metabolites, surfactants, and autoinducers. Sensing these bacterial products involves G protein-coupled receptors and nuclear hormone receptors. Moreover, DAMP-induced responses and ETI have emerged as important pathogen defense mechanisms in *C. elegans* (reviewed in [20]).

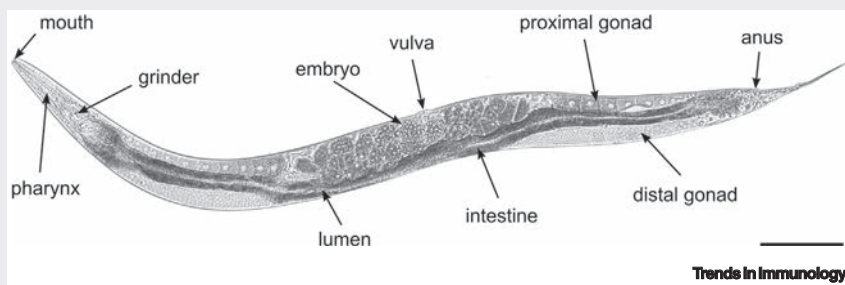


Figure 1. Light microscopic image of *C. elegans*. The gray bar represents 100 μ m. Courtesy of Dr. Barbara Pees.

provide a new perspective on the complexity and specificity of ETI.

There is a large body of research on defense responses to Bt PFTs in the *C. elegans* nematode host and several insect model species; this is because Bt PFTs are important in agriculture as pest control agents against insects and parasitic nematodes. Bt produces two types of PFTs effective against nematodes and insects: the three-domain (3d) Cry toxins (e.g., the nematocidal Cry5B, Cry12, Cry13, Cry14A, and Cry21A) and the alpha helical pesticidal App proteins (e.g., the nematocidal App6A and App6b) [11]. The cellular responses mounted by *C. elegans* after exposure to Cry and App PFTs have been characterized upon nematode exposure to either Bt spores that naturally harbor the PFTs [8], bacteria expressing heterologous PFTs [9], or purified PFT proteins [12]. The

observed defenses range from differential exocytosis, endocytosis, apoptosis, autophagy, protection via the interfilament network (associated with epithelia), to distinct humoral responses mediated, for example, by insulin-like signaling, various mitogen-activated protein kinase pathways, or C-type lectins [13].

By contrast to the consensus that pathogen effectors (which target common host processes) have similar functional outcomes (e.g., triggering the same innate immune pathways leading to expression of the same antimicrobial genes), our current understanding of the cellular response of *C. elegans* to Bt or Cry PFTs demonstrates astonishing specificity at two levels: specificity of the host response toward different PFTs (i) from different classes or (ii) from the same toxin class.

Specificity in the host response to PFTs from different classes

In the field of *C. elegans* immunity, the host responses to Cry or App PFTs are known to be distinct (Figure 1A,B). This specificity is remarkable because proteins from both classes form pores and thus target a common cellular function. However, Cry and App PFTs are structurally distinct and use different receptors to bind the host cell membrane. Thus, specificity may arise from differences in pore architecture, size, and permeability (e.g., permeability to Ca^{2+} or K^{+} ions), as well as from the interaction with various (often distinct) types of receptors in the target cell membrane.

Specificity in the host response to different PFTs from the same toxin class

Some *C. elegans* cellular defense response pathways are specifically required to defend against individual toxins within the same toxin class (Figure 1C–E), which is surprising because they share a common structure. For example, Bt679 kills nematodes through the production of two distinct 3d Cry PFTs: Cry21Aa3 and Cry14Aa2. The *C. elegans* GATA transcription factor ELT-2 mediates resistance to only one of the two (i.e., Cry21Aa; see Figure 1), demonstrating effector specificity of the host response [8]. Both toxins have common ancestry, which makes differences in the elicited host protective responses even more remarkable. Similarly, the *C. elegans* hypoxia-inducible transcription factor 1 mediates resistance to Cry5Ba but not Cry21Aa (Figure 1) [10]. These two Cry toxins use distinct host receptors to target the host cell membrane: Cry5Ba targets the nematode-specific cadherin CDH-8 [14] and glycolipid carbohydrates on the *C. elegans* cell membrane [15]; Cry21Aa targets the ABC transporter family protein ABT-4 [16].

Moreover, highly specific host responses are also expressed toward PFTs of the App class, which includes App6Aa and App6Ba. Specifically, App6Aa activates the

Key figure

Specificity in *Caenorhabditis elegans* effector-triggered immunity (ETI) to *Bacillus thuringiensis* (Bt) pore-forming toxins (PFTs)

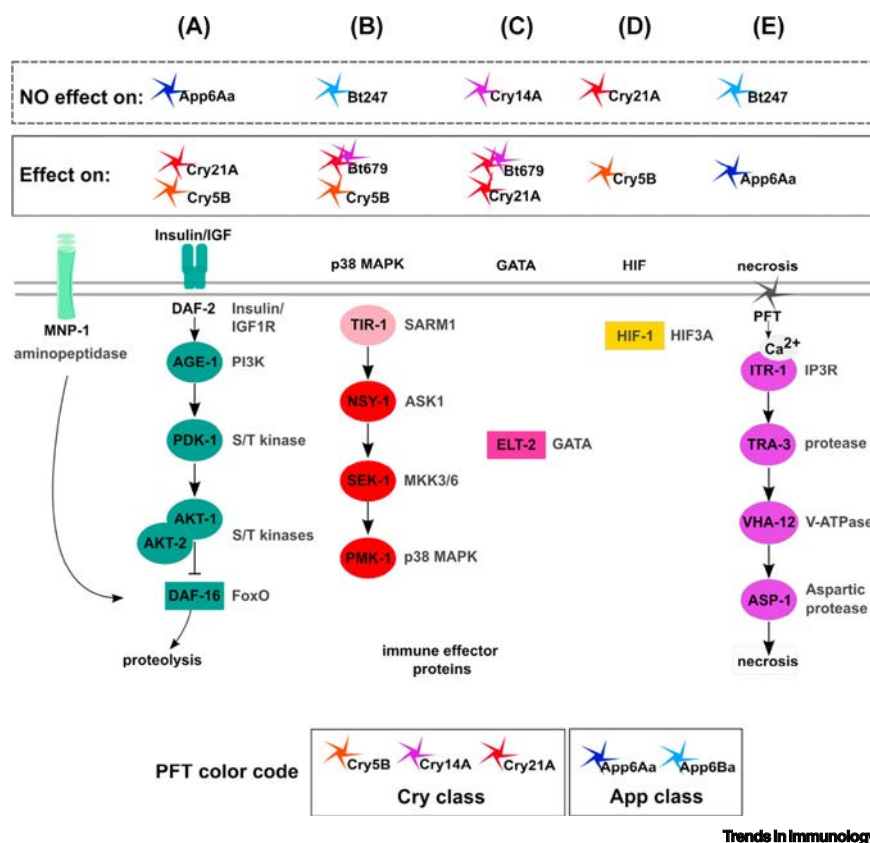


Figure 1. (A, B) Evidence of distinct defense responses to Cry and App PFTs is shown. (A) Knockout mutants of the intestinal transmembrane aminopeptidase N MNP-1 are hypersusceptible to Cry toxins Cry5B and Cry21A, but not to the App toxin App6Aa. MNP-1 activates proteolytic enzyme expression via insulin/insulin-like growth factor 1 (IGF-1) signaling, involving the FOXO transcription factor DAF-16 to promote survival during infection [6]. (B) *C. elegans* mutants of the p38 mitogen-activated protein kinase (MAPK) PMK-1 are hypersusceptible to exposure to the Cry toxin Cry5B and the Bt strain MYBt18679 (*Bt679*), which expresses the two Cry toxins Cry14A and Cry21A, but not to strain MYBt18247 (*Bt247*), which expresses the App toxin App6Ba [7,8]. (C–E) Evidence of distinct defense responses to PFTs from the same toxin class is shown. (C) Knockdown by RNAi of the intestinal GATA transcription factor gene *elt-2* renders *C. elegans* hypersusceptible to the Cry toxin Cry21A, but not to the Cry toxin Cry14A [8]. (D) Similarly, mutants of the hypoxia-inducible factor *hif-1* gene are hypersusceptible to the Cry toxin Cry5B, but not to the Cry toxin Cry21A [10]. (E) Aspartic protease *asp-1* necrosis-deficient mutants are tolerant to exposure to App6Aa, but *asp-1* mutants survive as well as wild-type worms when infected with Bt247, which expresses App6Ba [8,9]. Note: *asp-1* knockout has no effect on the susceptibility of *C. elegans* to the Cry toxin Cry5B [16] or the Cry toxin-producing Bt679 [8]. Abbreviations: ASK1, apoptosis signal-regulating kinase 1 (MAPKKK); ASP-1, aspartic protease 1; FOXO, Forkhead box O transcription factor; GATA, GATA transcription factor; insulin/IGF1R, insulin/insulin-like growth factor-1 receptor; IP3R, inositol 1;4;5-trisphosphate receptor; MKK, mitogen-activated protein kinase kinase; PI3K, phosphatidylinositol-3-OH kinase; SARM1, sterile alpha and TIR motif containing 1; (S/T) kinase, serine/threonine (S/T) kinase; V-ATPase, vacuolar-type ATPase.

host's endogenous necrosis pathway by directly interacting with the aspartic protease ASP-1. Activation of the necrosis pathway negatively influences host survival: *C. elegans asp-1* mutants are resistant to App6Aa exposure [9]. By contrast, *asp-1* animals are susceptible to infection with Bt247, which expresses App6Ba [8]. This remarkable specificity bares evidence of more complex effector-triggered innate immune responses than previously anticipated.

The *C. elegans* response to microbial effectors has been known for some time to show at least some level of specificity. For example, pore formation activates a membrane repair response, whereas inhibition of mitochondrial protein synthesis by bacterial effectors triggers the **mitochondrial unfolded protein response** [12,17]. However, the high level of specificity described here is still surprising, because all PFTs – regardless of differences in molecular structure – should cause comparable cellular damage; they all penetrate the target cell membrane, generate pores, and perturb cellular ion homeostasis. Because of these similarities, cellular responses to PFTs in vertebrates are usually considered to be rather general [1,5]. On the basis of information on Bt PFT-*C. elegans* interactions, we propose to rectify our view of PFTs and their presumably identical – or at least highly similar – effects on the host. PFTs harbor highly diverse mechanisms of action, pore architecture, size, and ion permeability and interact with various, often distinct types of receptors in the target cell membrane [3,4,14–16,18]. The combination of these PFT characteristics has the potential to elicit unique cellular responses toward different PFTs. Overall, it is largely unknown how PFT-specific responses are triggered, thus representing an exciting area for future research. Potential mechanisms may include direct sensing of the toxin (equivalent to recognition of a **pathogen-associated molecular pattern**), the pore structure, or the effects of membrane disruption. In this context, it

is essential to characterize in more detail the mechanism(s) of action of various PFTs, the affected cellular processes in the host, the mechanism(s) of recognition of PFTs themselves and/or their cellular effects, and the host responses they elicit.

This includes understanding the cellular 'decision-making' process that integrates multiple context-sensing input signals to produce a given defense response. The nematode *C. elegans* may serve as an informative, experimentally tractable model that can stimulate related research on ETI complexity and specificity in vertebrates, including humans.

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Declaration of interests

The authors have no interests to declare.

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References

- Remick, B.C. *et al.* (2023) Effector-triggered immunity. *Annu. Rev. Immunol.* 41, 453–481
- Stuart, L.M. *et al.* (2013) Effector-triggered versus pattern-triggered immunity: how animals sense virulent pathogens. *Nat. Rev. Immunol.* 13, 199–206
- Peraro, M.D. and van der Goot, F.G. (2016) Pore-forming toxins: ancient, but never really out of fashion. *Nat. Rev. Microbiol.* 14, 77–92
- Brito, C. *et al.* (2018) Mechanisms protecting host cells against bacterial pore-forming toxins. *Cell. Mol. Life Sci.* 76, 1319–1339
- Lopes Fischer, N. *et al.* (2020) Effector-triggered immunity and pathogen sensing in metazoans. *Nat. Microbiol.* 5, 14–26
- Chen, F. *et al.* (2023) Aminopeptidase MNP-1 triggers intestine protease production by activating daf-16 nuclear location to degrade pore-forming toxins in *Caenorhabditis elegans*. *PLoS Pathog.* 19, e1011507
- Huffman, D.L. *et al.* (2004) Mitogen-activated protein kinase pathways defend against bacterial pore-forming toxins. *Proc. Natl. Acad. Sci. U. S. A.* 101, 10995–11000
- Zárate-Potes, A. *et al.* (2020) The *C. elegans* GATA transcription factor elt-2 mediates distinct transcriptional responses and opposite infection outcomes towards different *Bacillus thuringiensis* strains. *PLoS Pathog.* 16, e1008826
- Zhang, F. *et al.* (2016) *Bacillus thuringiensis* crystal protein Cry6Aa triggers *Caenorhabditis elegans* necrosis pathway mediated by aspartic protease (ASP-1). *PLoS Pathog.* 12, e1005389
- Bellier, A. *et al.* (2009) Hypoxia and the hypoxic response pathway protect against pore-forming toxins in *C. elegans*. *PLoS Pathog.* 5, e1000689
- Wei, J.-Z. *et al.* (2003) *Bacillus thuringiensis* crystal proteins that target nematodes. *Proc. Natl. Acad. Sci. U. S. A.* 100, 2760–2765
- Los, F.C.O. *et al.* (2011) RAB-5 and RAB-11-dependent vesicle-trafficking pathways are required for plasma membrane repair after attack by bacterial pore-forming toxin. *Cell Host Microbe* 9, 147–157
- Pinos, D. *et al.* (2021) Response mechanisms of invertebrates to *Bacillus thuringiensis* and its pesticidal proteins. *Microbiol. Mol. Biol. Rev.* 85, e00007–e00020
- Peng, D. *et al.* (2018) Nematode-specific cadherin CDH-8 acts as a receptor for Cry5B toxin in *Caenorhabditis elegans*. *Appl. Microbiol. Biotechnol.* 102, 3663–3673
- Griffitts, J.S. *et al.* (2005) Glycolipids as receptors for *Bacillus thuringiensis* crystal toxin. *Science* 307, 922–925
- García-Montelongo, M. *et al.* (2021) Use of RNAi as a preliminary tool for screening putative receptors of nematocidal toxins from *Bacillus thuringiensis*. *Arch. Microbiol.* 203, 1649–1656
- Pellegrino, M.W. *et al.* (2014) Mitochondrial UPR-regulated innate immunity provides resistance to pathogen infection. *Nature* 516, 414–417
- Jimenez, A.J. *et al.* (2014) ESCRT machinery is required for plasma membrane repair. *Science* 343, 1247136
- Dierking, K. *et al.* (2016) Antimicrobial effectors in the nematode *Caenorhabditis elegans*: an outgroup to the Arthropoda. *Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci.* 371, 20150299
- Tran, T.D. and Luallen, R.J. (2024) An organismal understanding of *C. elegans* innate immune responses, from pathogen recognition to multigenerational resistance. *Semin. Cell Dev. Biol.* 154, 77–84